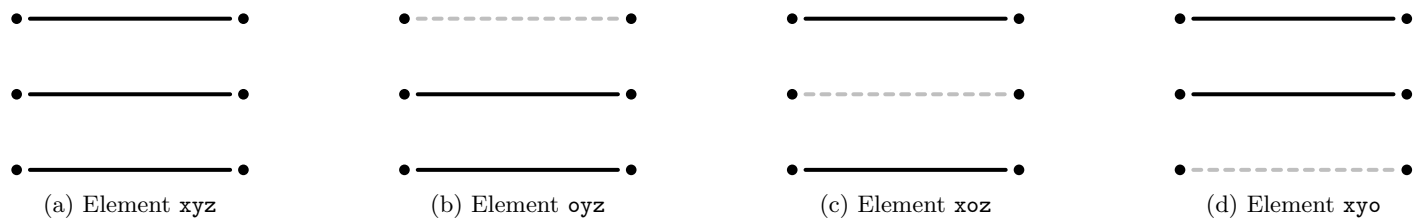


Algebra of color shufflings for cube faces

1 Generators



2 Extension operator $\otimes$

$$(l_1l_2l_3,m)\otimes(r_1r_2r_3,n)=(u_1u_2u_3,m+n)\qquad\text{where}\qquad m,n\in\mathbb{N}\qquad\text{and}\qquad u_i=\left\{\begin{array}{ll}\circ & \text{if }l_i=\circ\\r_j & \text{if }r_i=\circ\wedge l_j=\circ\\r_i & \text{if }l_i\neq\circ\neq r_i\end{array}\right.\qquad(1)$$

$\hookrightarrow$ has identity **xyz** but is not associative

3 Path weights

$$\mathcal{CF} \;=\; (\mathbb{N})\qquad\qquad\qquad(2)$$